

Technical Specifications: HackRF Pro

Radiofrequency Engineering

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1 Gains

Controlled separately through `libhackrf`:

- **LNA (IF RX):** `hackrf_set_lna_gain()` - 0 to 40 dB, 8 dB steps.
- **VGA (baseband RX):** `hackrf_set_vga_gain()` - 0 to 62 dB, 2 dB steps.
- **TX VGA:** `hackrf_set_txvga_gain()` - 0 to 47 dB, 1 dB steps.
- **AMP (RF front end):** `hackrf_set_amp_enable()` - 0 (off) or 1 (on, fixed $\approx +14$ dB). Applies to RX and TX.

Note: Adjust gain while avoiding ADC saturation; validate with the noise floor and absence of clipping.

2 Hardware Filters

- **Analog filter (Baseband LPF):** `hackrf_set_baseband_filter_bandwidth()`
Selectable thresholds: 1.75, 2.5, 3.5, 5, 5.5, 14, 20, 24, 28 MHz.
Transceiver: MAX2831 (Pro) - compatible with existing software through *legacy mode* in `libhackrf`.
- **Digital filtering and decimation (iCE40 FPGA):**
Configured implicitly according to the sample rate. Handles hardware decimation, removes the DC spike (previously handled in software), and enables 16-bit samples at low rates with a typical ENOB of 9–11 bits.

3 Frequency Ranges

- **Nominal range:** 100 kHz to 6 GHz.
- **Tunable range (physical limits):** 0 Hz to 7.1 GHz.
 - Below 100 kHz: analog sensitivity drops drastically.
 - Above 6 GHz: severe attenuation and unreliable calibration.

4 Sample Rate and Resolution

Mode	Resolution	Max. Sample Rate	Typical Use
Standard	8-bit I/Q	20 MS/s	General compatibility, up to 20 MHz of visible BW
Extended precision	16-bit	Low (FPGA decimation)	Narrowband signals, higher dynamic range (ENOB 9–11)
Medium precision	4-bit	40 MS/s	Wideband sweep; reduced resolution to fit over USB

5 IQ & DC Corrections

- **DC spike:** Removed in hardware by the iCE40 FPGA before sending data over USB. It no longer requires software filters (GNU Radio, etc.).
- **I/Q imbalance:**

Not documented in `libhackrf` or in the official firmware to date. Verify in future API or FPGA versions.

6 Connectors and Inputs

Connector	Function	Specification
USB-C	Power and data	USB 2.0 High Speed. 5 V / 500 mA min.
RF (SMA)	Main TX/RX	50 Ω . Half-duplex. Range: 100 kHz – 7.1 GHz.
P1 (SMA)	CLK IN (default)	10 MHz, 3.3 V square wave. Do not exceed 3.3 V or go below 0 V.
P2 (SMA)	CLK OUT (default)	10 MHz, 3.3 V. Daisy chain between devices.
P20	GPIO / ADC / RTC	22 pins, 3.3 V. GPIO, ADC, and auxiliary power access.
P22	I2S / SPI / I2C / UART	26 pins. Alternate CLK IN (enable with <code>hackrf_clock -c p22</code>).
P28	SD / Trigger	22 pins. SDIO, trigger IN/OUT for synchronization.

- **Integrated TCXO:** 0.5 ppm. Eliminates frequency drift without an external reference.
- **External clock:** Connect 10 MHz to P1. The hardware switches automatically when TX/RX starts. Verify detection with `hackrf_clock -i`.
- **Clock out:** Enable with `hackrf_clock -o 1`. Allows daisy chaining multiple HackRF Pro units.
- **RF Bias-T:** 3.3 V, 50 mA max. Software-enabled to power external LNAs.
- **GPIO:** All headers operate at 3.3 V CMOS. P22 includes I2C and UART for external peripheral control.
- **Note:** The audio jack and MicroSD slot are part of the **PortaPack H2** module, not the base HackRF Pro.

7 Safe Operating Limits

Critical warnings - read before connecting external signals

- **Maximum RX power:** Do not exceed **-5 dBm** to avoid distortion. Irreversible physical damage starts at **+10 dBm** (even with AMP disabled, a software error can enable it).
- **TX→RX loopback:** Use sufficient attenuation. The TX level can reach 10–15 dBm; without attenuation the front end is damaged.

- **Unloaded RF port:** Always terminate with an antenna or **50 Ω** during transmission.
- **External clock P1:** 10 MHz signal strictly between 0 V and 3.3 V. Do not use other frequencies except with firmware modification.
- **GPIO:** Do not exceed 3.3 V on any pin of headers P20/P22/P28.
- **ESD:** Handle with antistatic protection. Do not allow metal contact with the powered PCB.

8 Clock and Synchronization

- **Internal clock:** 0.5 ppm TCXO, active by default. Requires no configuration.
- **External clock:** 10 MHz signal on P1 (SMA). The device switches automatically when a TX/RX operation starts. Verify with `hackrf_clock -i`.
- **Clock out:** 10 MHz signal available on P2. Enable with `hackrf_clock -o 1`. Useful for daisy chaining units or synchronizing with laboratory instruments.
- **Alternate signals on P1/P2:** Using `hackrf_clock -1` or `hackrf_clock -2`, different internal signals can be connected instead of the defaults (trigger, FPGA clocks, etc.).
- **Alternate CLK IN:** P22 pin 2 exposes a second independent CLKIN input. Enable with `hackrf_clock -c p22`. Do not connect P1 and P22 pin 2 simultaneously as inputs.
- **Trigger IN/OUT:** Available on P28 pins 15 and 16. Allows capture-start synchronization between devices.
- **Multi-HackRF:** Connect CLKOUT from one unit to CLKIN of the next. Enable CLK-OUT with `hackrf_clock -o 1` before starting operation.

9 Indicator LEDs

When connecting the HackRF Pro to USB, **four LEDs should turn on:** MCU, FPGA, RF, and USB. If any of them does not turn on, it indicates a fault in the internal power chain or in the firmware.

LED	Function	Normal state
MCU	Primary internal power and active firmware	On when USB is connected
FPGA	Active iCE40 FPGA power	On when USB is connected
RF	Active RF-chain power	On when USB is connected
USB	USB communication established with host	On when USB is connected
RX	Reception operation in progress	On only during active RX
TX	Transmission operation in progress	On only during active TX

Note: LED colors have no meaning - they differ from one another to make visual distinction easier, not to indicate states.

10 Expansion Pinout (P20 / P22 / P28)

The HackRF Pro exposes three internal expansion headers. All operate at **3.3 V CMOS**.

P20 - GPIO / ADC / Power (22 pins)

Pin	Signal	Pin	Signal
1	VBAT	2	PB_5
3	3V3AUX	4	WAKEUP
5	GPIO3_8	6	GPIO3_9
7	GPIO3_10	8	GPIO3_11
9	GPIO3_12	10	GPIO3_13
11	GPIO3_14	12	GPIO3_15
13	GND	14	ADC0_6
15	GND	16	ADC0_2
17	VBUSCTRL	18	ADC0_5
19	GND	20	ADC0_0
21	VBUS	22	VIN

P22 - I2S / SPI / I2C / UART (26 pins, selection)

Includes: alternate CLKIN (pin 2), I2C1 (pins 5–6), UART (pins 19–20), I2S TX/RX, and FPGA auxiliary signals. See the full documentation at hackrf.readthedocs.io.

P28 - SD / Trigger / GPIO (22 pins)

Includes: full SDIO (pins 3–11), Trigger OUT (pin 15), and Trigger IN (pin 16) for hardware synchronization between devices. See the full documentation at hackrf.readthedocs.io.

Usage note: Do not exceed 3.3 V on any pin. Confirm maximum current per pin in the LPC43xx datasheet before connecting external loads.